

7	(a) minimum frequency of e.m. radiation/a photon (not “light”) for emission of electrons from a surface <i>(reference to light/UV rather than e.m. radiation, allow 1/2)</i>	M1	
		A1	[2]

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- (b) E_{MAX} corresponds to electron emitted from surface
 electron (below surface) requires energy to bring it to surface, so less than E_{MAX} B1 [2]
 B1
- (c) (i) $1/\lambda_0 = 1.85 \times 10^6$ (allow 1.82 to 1.88) C1
- $$f_0 = c/\lambda_0$$
- $$= 3.00 \times 10^8 \times 1.85 \times 10^6$$
- $$= 5.55 \times 10^{14} \text{ Hz}$$
- A1 [2]
- (ii) $\Phi = hf_0$
 $= 6.63 \times 10^{-34} \times 5.55 \times 10^{14}$ (allow ECF from (c)(i)) C1
 $= 3.68 \times 10^{-19} \text{ J}$ A1 [2]
- (d) sketch: straight line with same gradient
 intercept between 1.0 and 1.5 M1
 A1 [2]